National University of Computer and Emerging Sciences, Lahore Campus

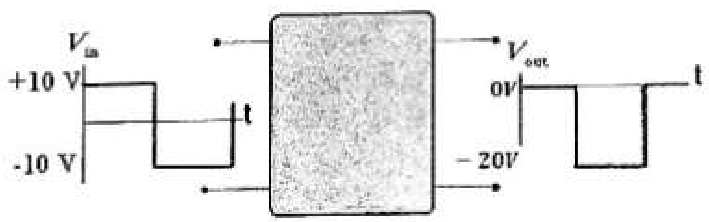
Course:	Electronic Devices and Circuits	Course Code:	EE225
Program:	BS(EE)	Semester:	Fall 2016
Duration:	180 Minutes	Total Marks:	100
Paper Date:	23rd December, 2016	Weight	50%
Section:	ALL	Page(s):	1
Exam:	Final	Roll No:	

Instruction/Notes: **Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together. State your valid assumptions clearly if you have to make any.**

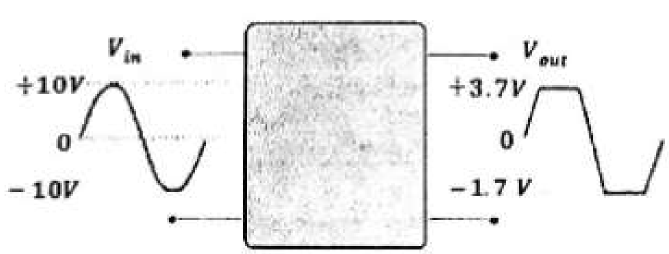
Question # 1 [3+2+2+2+3+2+2+3+3+3=25 marks]

Short Questions:

1) Draw the block diagram of a dc power supply.
2) Define threshold voltage in context of a MOS transistor?
3) What is Early voltage?
4) Explain channel pinch off in a MOS transistor?
5) Using a PNP BJT and voltage sources, draw a circuit in common base configuration such that it works in active mode.
6) What is the significance of using emitter resistance in a common emitter amplifier?
7) Why coupling capacitors are used in voltage amplifiers?
8) Draw the circuit inside the black box whose input and output are given below.



9) Draw the circuit inside the black box whose input and output are given below.



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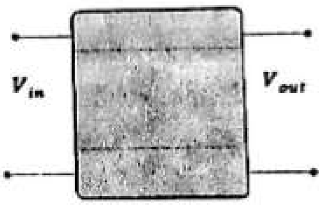
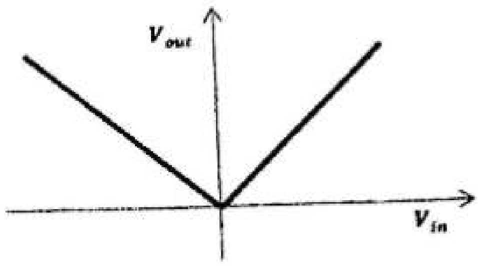
10) Draw the circuit inside the black box which has the transfer characteristics as given below.





Question # 2 [15 marks]

A shunt regulator utilizes a Zener diode whose voltage is 5.1V at a current of 50mA and whose incremental resistance is 7Ω. The diode is fed from a supply of 15V nominal voltage through a 200Ω resistor. What is the output voltage at no load? Find the line regulation and load regulation?

Question # 3 [10 marks]

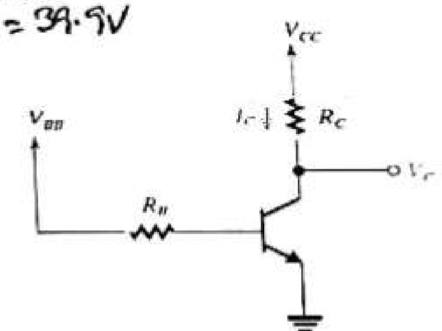
For the circuit shown in figure below, $V_{CC} = 10V, R_C = 10K\Omega$ and $R_B = 200K\Omega$. The BJT has $\beta = 100$. Find the value of V_{BB} that result in transistor operating

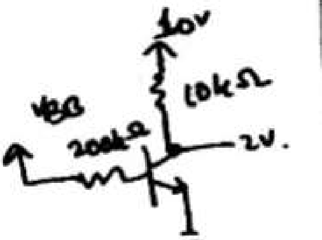
a) In active mode with $V_C = 2V$ $V_{BB} = 2.3V$

b) At the edge of saturation $V_{BB} = 2.64V$.

c) Deep in saturation with $\beta_{forced} = 20$.

$V_{BB} = 39.9V$





$V_E = 0$

$V_B = 0.7V$

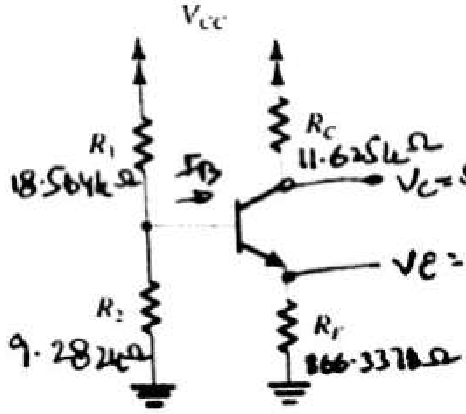
$\frac{10-2}{10k} = I$

I_C

$I_c (EOS) = \frac{V_{cc}}{R}$

Question # 4 [15 marks]

Design the circuit shown in the figure with β = 120 such that the transistor operates at a Q point of V_{CE} = 5V for V_{CC} = 15V, I_C = 0.8mA, and the voltage across resistor R_E is 0.7V. Assume the current flowing through potential divider network is equal to the emitter current.



$I = 0.8mA$

$I_E = 8.08$

$V_{BB} = 5V$

$R_{BB} = 6.546k\Omega$.

$I_B = 6.67\mu A$

$I_{AB} = I_E$.

GrcSU=AV.

Question # 5 [15 marks]

A common base amplifier is biased at $I_E = 0.25mA$ with $R_C = R_L = 10k\Omega$ and is driven by a signal source $R_{sig} = 1k\Omega$. Find the overall voltage gain G_v. If the maximum signal amplitude of the voltage between base and emitter is limited to $10mV$, what are the corresponding amplitudes of v_{sig} *and* v_o? Assume $\alpha \simeq 1$.

Question # 6 [10+10=20 marks]

a) An NMOS transistor operating in the triode region with $v_{DS} = 0.1V$, is found to conduct $60\mu A$ for $v_{GS} = 2V$ and $160\mu A$ for $v_{GS} = 4V$.

i. What is the apparent value of threshold voltage V_t?

ii. If $k_n' = 50\mu A/V^2$, What is the device W/L ratio?

iii. What current would you expect to flow with $v_{GS} = 3V$ and $v_{DS} = 0.15V$?

iv. If the device is operated at $v_{GS} = 3V$, at what value of v_{DS} will the drain end of MOSFET channel just reach pinch off, and what is the corresponding drain current?

b) In the circuit shown in figure, transistor is characterized by $|V_t| = 2V$, $k_n'W/L = 1mA/V^2$. Find the labeled voltages V_1 *and* V_2.